Strategyproof Facility Location with Interval Predictions and Feasibility Constraints

CS 6001 Course Project — Proposal

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1 Motivation and Problem Statement

In real-world infrastructure planning, whether placing a campus router, a transit stop, or a charging station, two standard assumptions of theoretical mechanism design frequently break down. First, machine learning models rarely output a perfectly certain single point; instead, they output confidence intervals reflecting epistemic uncertainty (e.g., "The optimal location lies somewhere in the range $[4.2, 5.8]$"). Second, the physical environment is rarely a continuous, unconstrained line; there are "blocked zones" such as staircases, restricted areas, or private property where placing a facility is physically impossible.

Existing learning-augmented mechanisms achieve strategyproofness by carefully projecting a single-point prediction onto a bounding interval constructed from the agents' reported locations. However, introducing inaccessible regions breaks this standard fallback mechanism. If the mechanism simply snaps the facility to the nearest valid physical location, it introduces a discontinuity that agents can exploit, instantly destroying strategyproofness. Furthermore, upgrading the ML hint from a point prediction to an interval prediction fundamentally changes how the mechanism must calculate "trust" in the machine-learned advice. Our project aims to answer: How can we design a learning-augmented mechanism that maintains strategyproofness on a constrained line using interval predictions, and how do environmental constraints impact the consistency-robustness tradeoff?

2 Related Work

Foundational work by Moulin (1980) established the median voter rule as strategyproof and optimal for minimizing the sum of costs on a line. Procaccia and Tennenholtz (2009) extended this by proving that for the maximum cost (egalitarian) objective, a 2-approximation is the best possible bound for any deterministic strategyproof mechanism.

Recently, Agrawal et al. (2022/2024) introduced a learning-augmented framework, demonstrating that mechanisms provided with a single-point prediction ($\hat{o}$) can bypass these worst-case lower bounds. Their fallback mechanism achieves 1-consistency while maintaining 2-robustness. While a comprehensive survey by Chan et al. (2021) covers numerous variants of facility location, including constrained facilities and obnoxious locations, the specific intersection of interval-based

ML predictions and environmental feasibility constraints remains unexplored in the strategyproof learning-augmented setting.

3 Proposed Approach

We propose a computational and experimental extension over the base mechanism by Agrawal et al. Our goal is to design, implement, and empirically evaluate a learning-augmented mechanism that maintains strategyproofness on a constrained line using interval predictions. Our approach relies on four concrete deliverables:

1. **The Discretized Environment:** A Python simulation utilizing NumPy to model a 1D grid containing predefined feasible and infeasible intervals, alongside synthetic agents drawing their true peaks from controlled distributions.
2. **The Mechanism:** An adaptation of the Agrawal fallback mechanism that accepts a confidence interval prediction $[p_{low}, p_{high}]$ and mathematically ensures the output facility location strictly falls within a feasible zone without introducing manipulative discontinuities.
3. **The Strategyproofness Validator:** A brute-force deviation checker that loops over all agents and all possible misreports on the constrained grid to computationally verify that our adaptation does not violate truthfulness.
4. **The Evaluation:** A quantitative analysis plotting the empirical approximation ratio (for both social and max cost) against two variables: the width of the ML prediction interval, and the density of the blocked zones.

4 Expected Timeline

- **Week 1–2:** Set up the discretized Python/NumPy environment. Implement the baseline median mechanism and the brute-force strategyproofness validator.
- **Week 3–4:** Implement the standard Agrawal single-point prediction mechanism. Develop and code our proposed extension to handle interval predictions and blocked zones.
- **Week 5:** Run large-scale simulations across varying prediction errors and blocked-zone densities. Generate consistency-robustness curves.
- **Week 6–7:** Run adversarial search for worst-case instances to test empirical robustness. Analyze data, finalize plots, and draft the 6-8 page final report.

References

[1] P. Agrawal, E. Balkanski, V. Gkatzelis, T. Ou and X. Tan, *Learning-augmented mechanism design: leveraging predictions for facility location*, Mathematics of Operations Research, 49(4):2626-2651, 2024.

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[3] H. Moulin, *On strategy-proofness and single peakedness*, Public Choice, 35(4):437-455, 1980.

[4] A. D. Procaccia and M. Tennenholtz, *Approximate mechanism design without money*, ACM Transactions on Economics and Computation, 1(4):18, 2013.